RCS-VRF — Circuit Structure, Entropy, and Certification

A primer on Leg 3 (circuits), where the entropy actually enters, and how Leg 5 certifies it

Design note · Legs 3–6 · for internal alignment

The four claims the protocol should deliver

1. **Statistical near-uniformity** of the output bits—indistinguishable from uniform to a polynomial-time adversary.
2. **Quantum origin**—the randomness comes from physical measurements, not classical computation.
3. **Adversary-robust**—claims 1 and 2 hold even if the quantum server is dishonest.
4. **Publicly verifiable**—anyone with the audit log can re-derive all numerical claims.

The seven legs at a glance

- **Leg 1 — Seed generation.** Produces an unpredictable, unbiasable, verifiable seed for each round. Multi-source hash-based commit–reveal (drand / NIST / BABE etc.), where unpredictability depends only on the hash and at least one honest source—no secret key.
- **Leg 2 — Parameter precommitment.** Locks circuit-generation parameters before the seed is used. The PRF/PRG step that deterministically expands the seed into the byte stream used downstream—expands, doesn't add entropy.
- **Leg 3 — Circuit generation.** Deterministically expands the seed into random circuits via SHAKE-256. Builds the challenge circuit ensemble (the edge-coloured $U_{ZZ} + SU(2)$, $d = 10$ family), providing the statistical structure—approximately Porter–Thomas output statistics and classical-simulation hardness—not entropy.
- **Leg 4 — Quantum sampling:** the QPU runs the circuits and returns bitstrings, *fast*—each sample must come back within a strict time bound, which is what precludes the adversary from simulating the circuits classically. **The sole source of certified entropy.** Vendors (IonQ, Quantinuum) operate here as untrusted miners.
- **Leg 5 — Verification:** computes $\mathcal{F}_{\mathrm{XEB}}$, derives $Q_{\min}$, and certifies the smooth min-entropy (the 71,313-bit result). The certification machinery.
- **Leg 6 — Extraction:** Toeplitz extractor converts certified entropy into uniform certified random bits.
- **Leg 7 — Anchoring:** makes the run verifiable after the fact by anchoring the audit trail (e.g. hash-graph). *(Exact mechanism yet to be decided.)*

This primer focuses on **Legs 3–6**, where the actual certified quantum entropy is set up, injected, and certified.

Leg 1 in brief — what we are going with

After some back and forth with Mehdi and incorporating his suggestions, what we have come up with so far is: the seed would be built from a **hash-based, multi-source canonical transcript**, not from any VRF per se. Its unpredictability rests on a hash function plus the assumption that *at least one* source is honest—there is no secret key whose secrecy the seed depends on. Four components combine, and defeating the seed requires defeating *all* of them:

- **Commit–reveal** from operator and public pool—no party can choose its contribution after seeing others'.
- **External beacons** drand and NIST, two independently-operated public pulses.

- **Hash-based public puzzle** for permissionless liveness (anyone can solve it).
- **Chained transcript** (`prev_finalized_round_hash`) linking each round to the last, making history tamper-evident.

We are *not* going with EC-VRF / Chainlink VRF. The problems with that route:

- **Secret-key dependence.** A VRF's unpredictability depends on a secret key; security collapses to trusting whoever holds it.
- **Single point of trust.** Chainlink is single-keyholder—one party can predict or steer every output.
- **MEV / withholding window.** Its request-then-fulfill workflow opens a gap between request and response in which the fulfiller can manipulate or withhold.
- **Not quantum-secure.** EC-VRF rests on elliptic-curve hardness, which a quantum adversary breaks.
- **The threshold-VRF fix has its own cost.** Distributing the key (BABE / drand-style) removes the single point of trust but only by operating a validator network—justifiable only when that network already exists for another reason.

What the hash-based, multi-source design (developed with Mehdi) solves:

- **No secret key** on the unpredictability path—so no keyholder to trust, removing the first three problems at once.
- **No network to operate.** Multi-source hashing gives the robustness of the threshold fix without standing up a validator set; we *consume* drand / NIST / BABE rather than run them.
- **Quantum-secure seed.** Unpredictability rests on a hash function, not elliptic-curve hardness.
- **Trust minimised to "any one honest source."** Defeating the seed requires defeating all sources at once.

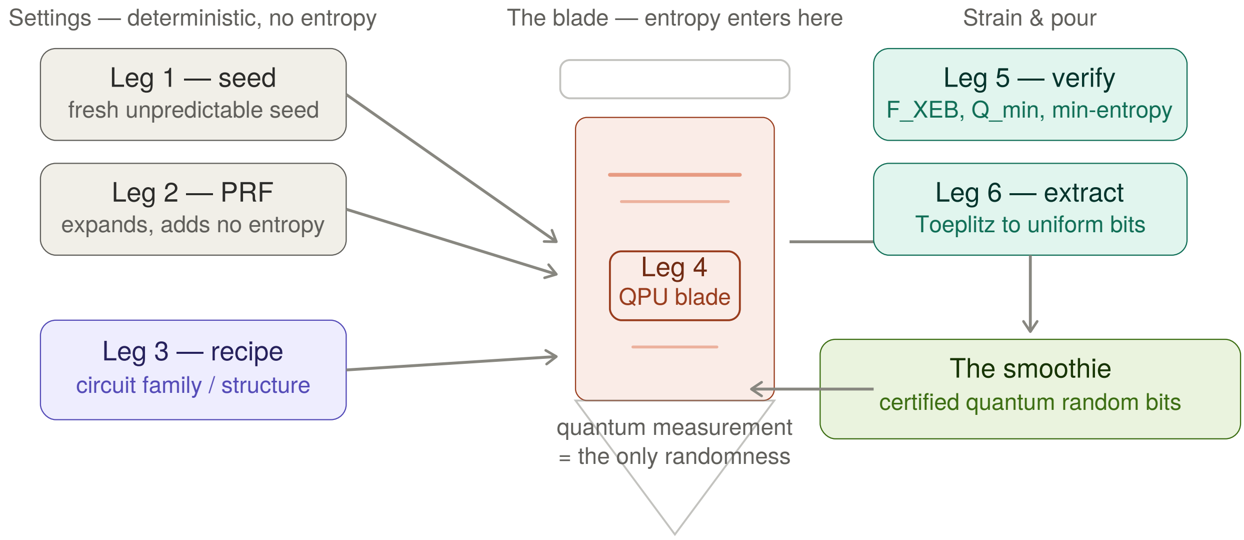
(Signing keys remain, but only for *attribution* / non-repudiation, not for unpredictability.) See the companion Leg 1 seed primer for the full rationale.

The shape of the whole thing

Three facts to hold onto before any detail:

- **Legs 1–3 add no entropy.** Given the seed, everything through circuit generation is deterministic. Their job is to *constrain the adversary* (make the circuits hard to simulate in the time budget) and *set up the statistical test*—not to produce randomness.
- **Entropy enters *only* at Leg 4.** The certified randomness is the measurement outcome of a quantum state. That is the sole injection point.
- **Legs 5–6 measure and refine it.** Leg 5 certifies how much entropy the measurements must contain; Leg 6 extracts it into uniform bits.

So "where does the randomness come from?" has exactly one answer: the quantum measurement in Leg 4. Everything else is certification machinery.



The protocol as a blender: Legs 1–2 are dials and Leg 3 is the recipe (all deterministic, no entropy); Leg 4 is the spinning blade where the quantum measurement creates the only randomness; Legs 5–6 strain and pour the certified bits. Leg 7 (not shown) seals the bottle for audit.

Leg 3 — the circuit family (statistical structure, not entropy)

Leg 3 builds the challenge circuits. Each is the same fixed topology with freshly-selected single-qubit gates: $d = 10$ layers of entangling $U_{ZZ}(\pi/2)$ gates, each sandwiched by layers of random $SU(2)$ gates on all $n = 56$ qubits; the two-qubit arrangement comes from edge-colouring a regular graph. The family does *two* jobs, easy to conflate:

- **Hardness.** The circuits are deep enough that no classical adversary can simulate them within the per-sample time budget. This is what *forces* an honest server to actually run a QPU—it is the security-relevant half.
- **Statistical structure.** The circuits produce output statistics close to **Porter–Thomas**, $f(p) = Ne^{-Np}$ with $N = 2^n$. This is what makes the verification test ($\mathcal{F}_{\text{XEB}}$) meaningful and gives each sample its entropy value.

How Porter–Thomas arises, and where t-designs fit

Think of Porter–Thomas as the *destination*—the specific distribution of bitstring probabilities you see at measurement—and approximate t-designs as the *vehicle*, the property a circuit acquires layer by layer on the way there.

The mechanism (the general picture). Porter–Thomas is the physical signature of global quantum scrambling. Two forces act together over the circuit's depth:

- **Local randomisation** — the continuous, non-Clifford $SU(2)$ rotations push each qubit's state off any discrete grid into the continuous Bloch sphere.
- **Global entanglement** — the U_{ZZ} entangling layers spread that local randomness across the whole register, intertwining the qubits.

As depth grows, the state vector performs a chaotic walk through the 2^n-dimensional Hilbert space; the more it scrambles, the more its output probabilities settle toward the exponential law $f(p) = Ne^{-Np}$.

Where t-designs come in. An exact Haar-random state is the idealised limit—it would take infinitely deep circuits to reach perfectly. t-designs are the *moment-by-moment* milestones that track how far along that road a finite circuit has travelled: an ensemble is an approximate t-design if it matches the Haar measure's first t moments. The first milestone that matters here is the **2-design** (anti-concentration): the second moment governs the spread of the distribution, and reaching it means probability has flattened globally so no single bitstring dominates. Higher t would mimic higher moments, edging closer to full Haar.

Where our family actually sits — and why that is enough. We do get Porter–Thomas-like output statistics, and in that sense the family is often described as approaching Haar-random—"Haar-like" *(yet to be verified rigorously; this is our interpretive reading, not a claim the paper itself makes, and depth-10 likely precludes it at high moment-orders)*. Stated precisely: the $d = 10$ family does *not* travel the whole road. At depth 10 on 56 qubits the ensemble is provably *not* a high-t design (a lightcone leaves qubit pairs uncorrelated that Haar would correlate), and the state is not fully Haar-scrambled. What the family does is reach the *low moments the certification actually reads*—the $t = 2$ behaviour the $\mathcal{F}_{\mathrm{XEB}}$ mean depends on and the $t = 3$ behaviour its variance depends on—while remaining far from Haar everywhere above. That is the whole design intent: the protocol's test is structurally blind beyond $t \approx 3$, so a shallow circuit that matches those low moments produces output indistinguishable, *to the test*, from a truly Haar-random one—at a depth the QPU can run with high fidelity. The family *exhibits* approximately Porter–Thomas output statistics without being Haar-like as an ensemble; the implication Haar $\Rightarrow$ Porter–Thomas runs one way only, and "produces Porter–Thomas" never licenses "is Haar-like."

Two further notes for defensibility. First, this property is a property of the circuit *structure* (depth, gate set, two-qubit gate, topology)—not of the PRF that feeds it. The PRF (Leg 2) supplies uniform selection bits; a good PRF lets the family reach its structural ceiling but cannot raise it. Second, the protocol does not *derive* Porter–Thomas from a design theorem—it *posits* it and verifies it *numerically* (total-variation distance to Porter–Thomas and Shannon-entropy agreement, both decaying exponentially in n for this family). There is no t-design theorem in the chain; the low-moment agreement is checked, not proven.

What the paper actually states (Liu et al., SM Sec. III C 1)

The assume-and-verify structure is explicit in the source. The first set of circuit-sampling assumptions reads, verbatim:

> *"The output probabilities of any circuit C belonging to the circuit family follow the Porter–Thomas distribution given in Eq. III.4."*
>
> *"Achieving high XEB classically is as hard as computing the output probabilities of the quantum circuit for the chosen circuit family."*

Two things to read off this. (1) Porter–Thomas is listed as an *assumption*, not a derived result—and it sits *separately* from the hardness assumption, so neither flows from the other. (2) The justification the paper offers is numerical: it cites prior evidence that random-circuit outputs follow Porter–Thomas in general (Arute; Boixo), and for *this* family shows each circuit's distribution is close to Porter–Thomas in TVD (decaying exponentially in n) with Shannon entropy matching as well (Fig. S3). No t-design, no Haar argument, no derivation—assumed, then verified numerically.

Leg 4 — where the entropy actually enters

The QPU runs each challenge circuit once and returns one bitstring $x_i \in \{0, 1\}^n$. Because the circuit's output distribution is Porter–Thomas, a genuine measurement of it is unpredictable: the outcome cannot be computed from the seed or any classical side information. **This is the only step that creates entropy.** Vendors (IonQ, Quantinuum) operate here as *untrusted* miners—the whole point of certification is that we do not have to trust them.

Why this is a separate leg from Leg 3. It is tempting to fold the circuit family and the sampling into one "entropy" leg, but they do different things. Leg 3 is *deterministic given the seed*: the circuit C_i is a fully-reproducible *question*, with no entropy in it (anyone with the seed regenerates it exactly, the adversary included). The entropy is in the *answer*—the measurement outcome x_i, which the Born-rule collapse makes unpredictable even to someone holding the full circuit. The test for "is this leg a source of entropy?" is simply: *could someone with the seed reproduce its output?* For the circuit, yes (no entropy); for the bitstring, no (entropy). Leg 3 *enables* certified entropy by giving the measurement its Porter–Thomas structure; Leg 4 *is* the source. They are also where the security argument splits—Leg 3 supplies hardness (hard to simulate), Leg 4 supplies speed (returns fast)—so keeping them separate is what makes the adversary model legible.

Leg 5 — how certification is done

Certification answers: *given what came back, how much genuine quantum entropy must it contain?* It runs in three moves.

Step 1 — the $\mathcal{F}_{\text{XEB}}$ test

On a random subset $\mathcal{V}$ of $m = 1522$ circuit–sample pairs, compute

$$\mathcal{F}_{\text{XEB}} \;=\; \frac{N}{m}\sum_{i\in\mathcal{V}} p_{C_i}(x_i) \;-\; 1, \qquad p_{C_i}(x_i) = |\langle x_i|C_i|0\rangle|^2,$$

where the ideal probabilities $p_{C_i}(x_i)$ are computed by classical supercomputer simulation. Honest quantum samples concentrate $\mathcal{F}_{\text{XEB}}$ near 1; uniform (entropy-free) samples near 0. The protocol accepts only if $\mathcal{F}_{\text{XEB}} \geq \chi$ *and* the average time per sample is below threshold. In the reference run, $\mathcal{F}_{\text{XEB}} = 0.32$ against $\chi = 0.30$.

What the test can and cannot see. $\mathcal{F}_{\text{XEB}}$ is a *first-moment* statistic. Its *expectation* is fixed by $\sum_x p_x^2$ (the second moment of p, i.e. anti-concentration / $t = 2$); its *sampling fluctuations* reach $\sum_x p_x^3$ ($t = 3$). It is structurally blind to everything above $t = 3$ and to all long-range correlations—so it *cannot* certify full Haar-randomness, and does not try to. This is exactly why a depth-10 shortcut that matches those low moments passes it.

Step 2 — from the test to $Q_{\min}$

The security analysis models a restricted adversary that returns Q genuine quantum samples and spoofs the remaining $M - Q$ by finite-fidelity classical simulation (frugal rejection sampling). Bounding the probability that such an adversary clears $\mathcal{F}_{\text{XEB}} \geq \chi$ yields $\varepsilon_{\text{adv}}(Q, \chi)$, and inverting it gives

$$Q_{\min} \;=\; \min\{\, Q : \varepsilon_{\text{adv}}(Q, \chi) \geq \varepsilon_{\text{sou}} \,\},$$

the fewest quantum samples an adversary *must* have used to pass. In the reference run, at $\varepsilon_{\text{sou}} = 10^{-6}$ against an adversary $4\times$ Frontier, $Q_{\min} = 1297$.

Step 3 — from $Q_{\min}$ to certified entropy

Each genuine quantum round carries entropy equal to the *collision entropy* of the Porter–Thomas distribution:

$$H_2(X_i \mid C_i) \;=\; -\log_2 \textstyle\sum_x p_x^2 \;=\; -\log_2(2/N) \;=\; n - 1.$$

Note this $(n - 1)$ is the *same* $\sum_x p_x^2$ quantity the $\mathcal{F}_{\text{XEB}}$ mean reads—so "what the test checks" and "what each sample is worth" are one $t = 2$ object, not two assumptions. Multiplying through gives the certified smooth min-entropy

$$H_{\min}^{\varepsilon_s}(X^M \mid \tilde{I}_{\text{sn}}) \;\geq\; Q_{\min}\,(n-1) \;-\; \log\tfrac{1}{\varepsilon_s}.$$

In the reference run this is $H_{\min} \approx 71{,}313$ bits.

Leg 6 — extraction

The $56 \times 30{,}010$ raw bits, now certified to contain $\geq H_{\min}$ bits of min-entropy, are passed through a Toeplitz (quantum-proof strong) extractor, yielding $\approx 71{,}273$ near-uniform certified random bits. Extraction converts *certified entropy* into *uniform bits*; it neither adds nor certifies entropy itself.

The one caveat that governs every number

"Certified" always means relative to the restricted adversary model—bounded classical power $\mathcal{A}$, simulation-hardness assumptions, no postselection, one shot per circuit. The protocol does not prove quantumness unconditionally (that is the Bell paradigm, which it deliberately does not use). It proves: *under these assumptions*, a passing $\mathcal{F}_{\mathrm{XEB}}$ within the time bound could not have come from a bounded classical adversary, so the bits carry quantum-origin entropy. Every certified figure inherits that conditioning.

In summary

Legs 1–3 are all deterministic given the seed and contribute *zero* certified entropy—their job is to time-constrain the adversary and set up the statistical test. Entropy enters **only at Leg 4**. Legs 5–6 measure and refine it; Leg 7 makes it auditable.

Reference run (Liu et al., H2-1): $n = 56$, $M = 30{,}010$, $m = 1{,}522$, $\mathcal{F}_{\mathrm{XEB}} = 0.32$, $\chi = 0.30$, $Q_{\min} = 1{,}297$, $H_{\min} \approx 71{,}313$, extractable $\approx 71{,}273$, at $\varepsilon_{\mathrm{sou}} = 10^{-6}$, $\mathcal{A} = 4\times$ Frontier.